Rafael Consentino de la Vega

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Fields: Development Economics, Labour Economics, Structural Change, Income Inequality.

Education

United Nations University (UNU-MERIT) 2021–present
PhD Fellow, Economics (degree awarded by Maastricht University) Maastricht, NL
- *Essays on structural change and income inequality*. Expected submission: Feb/2027.
- Secondment at the University of Cape Town (Oct–Dec 2023).

Federal University of Rio de Janeiro (UFRJ) 2013–2015
MSc, Industrial Engineering Rio de Janeiro, BR

Federal University of Rio de Janeiro 2013–2018
BSc, Economics Rio de Janeiro, BR

Federal University of Rio de Janeiro 2007–2012
BSc, Industrial Engineering Rio de Janeiro, BR
- Exchange student at INSA Lyon, FR (2010–2011).

Research Experience

United Nations Industrial Development Organisation (UNIDO) 2025–present
Research Economist (currently part-time) Vienna, AT

UNU-MERIT 2023–2024
Research Assistant Maastricht, NL

Federal (UFRJ) & State (UERJ) Universities of Rio de Janeiro 2008–2017 / 2019–2021
Researcher Rio de Janeiro, BR

Teaching and Other Experience

UNU-MERIT / University College Maastricht 2022
Tutor in Development Economics Maastricht, NL

Federal University of Rio de Janeiro 2014–2017 / 2019–2021
Part-time lecturer in Economics and Engineering, Engineering BSc programmes Rio de Janeiro, BR

Federal (UFRJ) & State (UERJ) Universities of Rio de Janeiro 2017–2021
Part-time lecturer and coordinator, MBA in Industrial Engineering Rio de Janeiro, BR

Freelance 2013–2018
Patent translator, English and French to Portuguese Rio de Janeiro, BR

Job Market Paper

Automation and inequality within firms: Evidence from the manufacturing sector in South Africa
w/ T. Köhler, A. Martins-Neto, and S. Ngidi *UNU-WIDER WP Series, No. 61/2026* [Latest Draft]

Abstract. This paper investigates the effects of automation on employment dynamics and within-firm earnings inequality in South Africa's formal manufacturing sector. Leveraging matched employer-employee administrative records linked to firm-level import transactions, we find that automation causes a reduction in employment in automating firms relative to those that have not automated yet. However, broadening the control group to include firms that never automate, we find that automating firms are typically on a strong growth trajectory prior to adoption and continue to expand thereafter. This suggests that automation could be associated with a relative deceleration in employment growth rather than with job losses. Overall, average earnings show a modest decline post-automation, while within-firm inequality rises temporarily, although estimates are not always statistically significant. These findings suggest that automation may contribute to rising earnings dispersion within firms, especially in contexts of more substantial technological upgrading.

References

Tommaso Ciarli
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UNU-MERIT
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verspagen@merit.unu.edu

Alejandro Lavopa
UNIDO
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Research Grants and Awards

- IDB grant: AI firms, skills, and productivity in Brazil (18,000 USD; joint application).
- UNU-WIDER grant: economic structure and top inequality in South Africa (8,640 EUR; solo applicant).
- UNU-WIDER grant: automation and within-firm inequality in South Africa (4,500 EUR; solo applicant).
- Best lecturer of UFRJ Polytechnic School in 2017 (student vote).
- Fully-funded PhD fellowship, UNU-MERIT.
- Fully-funded Master's fellowship, Brazilian Federal Agency for Support and Evaluation of Graduate Education.

Languages, Skills, and Other Information

Citizenship: Brazilian, Italian (in process)
Languages: Portuguese (native), English (fluent), French (advanced), Spanish (intermediate), Italian (basic)
Software: Python, Stata, R, LaTeX
Data: RAIS and PNAD/C (Brazil); NT-SDF/SA-TIED firm-level tax data (South Africa)

Participation in Events

Organisation

- UNU-MERIT/YSI Workshop on Structural Change, Growth and Inequality, Maastricht 2025
- Summer School on Agent-Based Modelling in Economics and Social Sciences, Maastricht 2025

Conference presentations

- 2nd International Conference on Inequality and Opportunities, Bari 2026
- 3rd World Inequality Conference, Paris School of Economics, Paris 2026
- 3rd International Conf. on Regional Development in South America (KIRDSA), Buenos Aires 2026
- 11th Conference of the Society for the Study of Economic Inequality (ECINEQ), Washington, D.C. 2025
- 20th Doctorissimes – Paris School of Economics PhD Conference, Paris 2025
- SA-TIED Work-in-progress Workshop, Pretoria 2024
- 18th Meta-Analysis in Economics Research Network (MAER-Net) Colloquium, Palma de Mallorca 2023

Seminar presentations

- Department of Economics, Fluminense Federal University, Niterói (online) 2026
- Southern Centre for Inequality Studies, University of the Witwatersrand, Johannesburg (online) 2024
- School of Economics, University of Cape Town, Cape Town 2023
- UNU-MERIT Internal Conferences and Research Weeks, Maastricht 2021–2026

Summer schools

- 1st Summer School on Inequality and Opportunities, Bari 2026
- 3rd Codechella (advanced difference-in-differences and causal inference), Madrid 2026
- 25th ECLAC Development Policy Studies Program, Santiago 2024

Publications and Working Papers

Economic development and structural change

Working papers

- **de la Vega, R.**, Köhler, T., Martins-Neto, A., & Ngidi, S. (2026). Automation and inequality within firms: Evidence from the manufacturing sector in South Africa. *UNU-WIDER Working Paper Series*, No. 61/2026. (Job Market Paper). [Link]
- **de la Vega, R.** (2025). Economic structure and top earnings inequality in South Africa: A firm-level and sectoral perspective. *UNU-WIDER Working Paper Series*, No. 39/2025. [Link]
- **de la Vega, R.** (2023). Structural change and income inequality: A meta-analysis. *UNU-MERIT Working Paper Series*, #2023-046. [Link]

Work in progress

- **de la Vega, R.**, Jaccoud, F., Menéndez, M., Seri, C., & Ciarli, T. (forthcoming). Skills and AI innovative ecosystems: Evidence from Brazilian microregions. *IDB Working Paper Series*.
- Oliveira, A., Ciarli, T., **de la Vega, R.**, & Scholten, C. (forthcoming). The future of digital automation technologies: A real-time Delphi survey.

Policy reports

- Prettner, K., Bennett, N., Bloom, D., Hanna, T., **de la Vega, R.**, & Lavopa, A. (2026). The future of jobs in an era of demographic and technological transformation. *UNIDO Policy Brief Series*, #31. [Link]
- Proença, A., Cardoso, V., Caulliraux, H., Mallet, L., & **de la Vega, R.** (2013). Roadmap para o desenvolvimento do cluster de biotecnologia para a saúde humana do Estado do Rio de Janeiro [Roadmap for the development of the healthcare biotechnology cluster in Rio de Janeiro state]. Rio de Janeiro: SEBRAE. 36 p.

Operations management, health systems, and governance

Journal articles

- **de la Vega, R.**, Pinto, G., Ribeiro, F., & Spiegel, T. (2022). Pull processes in health care: A systematic literature review. *Gestão & Produção*, 29. [Link]
- **de la Vega, R.**, Klojda, T., Alexandre, R., & Spiegel, T. (2021). Análise por simulação de melhoria de processos na entrada de unidade de atenção primária de saúde [Simulation analysis of improvements in reception processes of a primary care unit]. *Revista Gestão & Saúde (Brasília)*, 12, 92–107. [Link]

Books

- Silva, E., Soares, P., **de la Vega, R.**, & Lacerda, D. (2017). *Inovação na Gestão Pública – A Construção da Empresa Bio-Manguinhos* [Innovation in public management – Building the Bio-Manguinhos State-Owned Enterprise]. Porto Alegre: Bookman. 351 p.